Section E

1. Consider $E[X] = -0.05$ to find unknown values from the following table. **[2]**

x	-2	-1	a	1	2
$P(X = x)$	0.2	0.25	0.1	b	0.15

$$\mathbf{0.2 + 0.25 + 0.1 + b + 0.15 = 1 \rightarrow b = 0.3}$$

$$\mathbf{(-2)(0.2) + (-1)(0.25) + a(0.1) + 1(0.3) + 2(2.05) = -0.05 \rightarrow -0.05 + 0.1a = -0.05 \rightarrow a = 0}$$

2. Let a random experiment be the casting of a pair of fair *four-sided* dice and let X equal the minimum of two outcomes. With reasonable assumptions, find the variance of X. **[3]**

$\boldsymbol{x}$	$\mathbf{1}$	$\mathbf{2}$	$\mathbf{3}$	$\mathbf{4}$
$\boldsymbol{P(X = x)}$	$\mathbf{\frac{7}{16}}$	$\mathbf{\frac{5}{16}}$	$\mathbf{\frac{3}{16}}$	$\mathbf{\frac{1}{16}}$

$$\boldsymbol{E(X) = 1 \times \frac{7}{16} + 2 \times \frac{5}{16} + 3 \times \frac{3}{16} + 4 \times \frac{1}{16} = 1.875}$$

$$\boldsymbol{E(X^2) = 1^2 \times \frac{7}{16} + 2^2 \times \frac{5}{16} + 3^2 \times \frac{3}{16} + 4^2 \times \frac{1}{16} = 4.375}$$

$$\boldsymbol{V(X) = 4.375 - 1.875^2 = 0.86}$$

3. A continuous random variable X has the *cdf* as

$$F(x) = \begin{cases} l_1; & x < 5 \\ \frac{x-5}{5}; & 5 \le x < 10 \\ l_2; & x \ge 10 \end{cases}$$

(a). What are the values of l_1 and l_2? **[1]**
(b). Estimate $P(7 \le X < 12)$. **[1]**
(c). Find the mean and the median of the distribution. **[3]**

(a) $\boldsymbol{l_1 = 0}$ **and** $\boldsymbol{l_2 = 1}$

(b) $\boldsymbol{P(7 \le X < 12) = F(12) - F(7) = 1 - \frac{2}{5} = 0.6}$

(c) $\boldsymbol{f(x) = F'(x) = \frac{1}{5}; 5 \le x \le 10}$

$$\boldsymbol{\mu = E(X) = \int_5^{10} x\left(\frac{1}{5}\right) dx = \frac{1}{5}\int_5^{10} x dx = \frac{1}{5}\left[\frac{x^2}{2}\right]_5^{10} = \frac{1}{10}(100 - 25) = 7.5}$$

$$\boldsymbol{F(m) = 0.5 \rightarrow \frac{m-5}{5} = 0.5 \rightarrow m = 7.5}$$

Section G

1. For $f(x) = \frac{x}{c};\ x = 1, 2, \dots \dots 10$, determine the constant c for which $f(x)$ is the *pmf* for a random variable X. **[1]**

$$\sum_{x=1}^{10} \frac{x}{c} = 1 \rightarrow \frac{1}{c}(1 + 2 + \cdots \cdots + 10) = 1 \rightarrow \frac{1}{c}\frac{10(10+1)}{2} = 1 \rightarrow \frac{55}{c} = 1 \rightarrow c = 55$$

$$f(x) = \frac{x}{55};\ x = 1, 2, \dots \dots 10$$

2. Let the random variable X have the *pmf* $f(x) = \frac{x}{10}; x = 1, 2, 3, 4$. Compute $P(X < 3)$, $E(2X^2 + 5)$, and $V(3)$. **[4]**

x	1	2	3	4
$P(X = x)$	$\frac{1}{10}$	$\frac{2}{10}$	$\frac{3}{10}$	$\frac{4}{10}$

$$P(X < 3) = \frac{1}{10} + \frac{2}{10} = 0.3$$

$$E(X) = 1 \times \frac{1}{10} + 2 \times \frac{2}{10} + 3 \times \frac{3}{10} + 4 \times \frac{4}{10} = 3$$

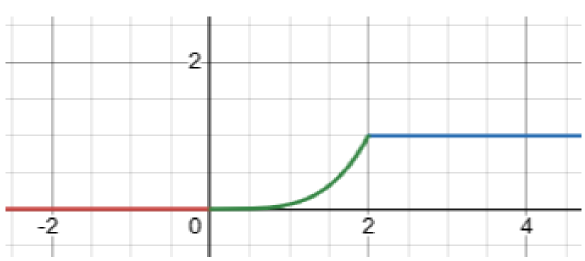
$$E(X^2) = 1^2 \times \frac{1}{10} + 2^2 \times \frac{2}{10} + 3^2 \times \frac{3}{10} + 4^2 \times \frac{4}{10} = 10$$

$$E(2X^2 + 5) = 2E(X^2) + 5 = 2 \times 10 + 5 = 25$$

$$V(3) = 0$$

3. Let X have the *pdf* $f(x) = \frac{x^3}{4}; 0 \leq x \leq 2$. Then, find and sketch the graph of *cdf*. Also, estimate the $P(X \geq 1)$ and median. **[5]**

$$F(x) = \int_0^x \frac{w^3}{4} dw = \left[\frac{w^4}{16}\right]_0^x = \frac{x^4}{16}; 0 \leq x \leq 2$$

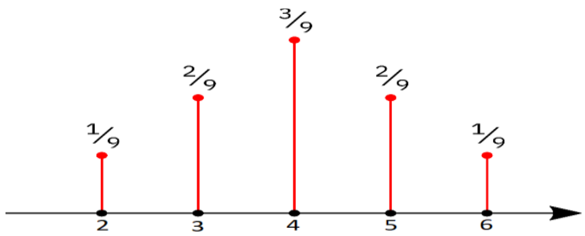


$$P(X \geq 1) = 1 - F(1) = 1 - \frac{1}{16} = \frac{15}{16}$$

$$F(m) = 0.5 \rightarrow \frac{m^4}{16} = 0.5 \rightarrow m = \sqrt[4]{8}$$

Section O

1. Line graph of a random variable X is given in the figure below. Compute $P(X \geq 3)$ and standard deviation. Also, find the median of X. **[5]**



$$P(X \geq 3) = \frac{2}{9} + \frac{3}{9} + \frac{2}{9} + \frac{1}{9} = \frac{8}{9}$$

x	2	3	4	5	6
$P(X = x)$	$\frac{1}{9}$	$\frac{2}{9}$	$\frac{3}{9}$	$\frac{2}{9}$	$\frac{1}{9}$

$$E(X) = 2 \times \frac{1}{9} + 3 \times \frac{2}{9} + 4 \times \frac{3}{9} + 5 \times \frac{2}{9} + 6 \times \frac{1}{9} = 4$$

$$E(X^2) = 2^2 \times \frac{1}{9} + 3^2 \times \frac{2}{9} + 4^2 \times \frac{3}{9} + 5^2 \times \frac{2}{9} + 6^2 \times \frac{1}{9} = 17.333$$

$$\sigma = \sqrt{17.333 - 4^2} = 1.155$$

$$\text{Median } X = 4$$

2. Let X have the *pdf* $f(x) = x - kx^3; 0 \leq x \leq 2$. Find the value of k and then estimate the $P(X < 1)$, $P(X \geq 1)$, and mode of X. **[5]**

$$\int_0^2 (x - kx^3)dx = 1 \rightarrow \left[\frac{x^2}{2} - \frac{kx^4}{4}\right]_0^2 = 1 \rightarrow 2 - 4k = 1 \rightarrow k = \frac{1}{4}$$

$$f(x) = x - \frac{x^3}{4}; 0 \leq x \leq 2$$

$$F(x) = \int_0^x \left(w - \frac{w^3}{4}\right) dw = \left[\frac{w^2}{2} - \frac{w^4}{16}\right]_0^x = \frac{x^2}{2} - \frac{x^4}{16};\ 0 \leq x \leq 2$$

$$P(X < 1) = F(1) = \frac{1}{2} - \frac{1}{16} = \frac{7}{16}$$

$$P(X \geq 1) = 1 - P(X < 1) = 1 - \frac{7}{16} = \frac{9}{16}$$

$$f'(x) = 1 - \frac{3}{4}x^2$$

$$f'(m_o) = 0 \rightarrow 1 - \frac{3}{4}m_o^2 = 0 \rightarrow m_o = \frac{2}{\sqrt{3}}$$

Section P

1. In a bet, the betting person wins \$1, \$2 and \$3 with probabilities 0.25, 0.2 and 0.15, and loses \$1 with probability 0.4 for each \$1 bet. Find $V(3-2X)$. **[3]**

$\boldsymbol{x}$	$\mathbf{1}$	$\mathbf{2}$	$\mathbf{3}$	$\mathbf{-1}$
$\boldsymbol{P(X=x)}$	$\mathbf{0.25}$	$\mathbf{0.2}$	$\mathbf{0.15}$	$\mathbf{0.4}$

$$\boldsymbol{E(X) = 1\times 0.25 + 2\times 0.2 + 3\times 0.15 + (-1)\times 0.4 = 0.7}$$

$$\boldsymbol{E(X^2) = 1^2\times 0.25 + 2^2\times 0.2 + 3^2\times 0.15 + (-1)^2\times 0.4 = 2.8}$$

$$\boldsymbol{V(3-2X) = 4V(X) = 4\{2.8 - 0.7^2\} = 9.24}$$

2. Let a random experiment be the casting of a pair of fair *four-sided* dice and let X equal the difference of two outcomes. With reasonable assumptions construct the probability distribution table. **[2]**

$$\boldsymbol{P(X=0) = P\{(1,1),(2,2),(3,3),(4,4)\} = \tfrac{4}{16}}$$

$$\boldsymbol{P(X=1) = P\{(1,2)(2,3),(3,4),(4,3),(3,2),(2,1),\} = \tfrac{6}{16}}$$

$$\boldsymbol{P(X=2) = P\{(1,3),(2,4),(4,2),(3,1)\} = \tfrac{4}{6}}$$

$$\boldsymbol{P(X=3) = P\{(1,4),(4,1)\} = \tfrac{2}{16}}$$

3. Let $f(y) = \frac{3}{2}y^2$; $-1 < y < 1$ be the *pdf* of a continuous random variable Y. Find mean, $P(Y>0)$, and Q_1 of Y. **[5]**

$$\boldsymbol{\mu = E(X) = \int_{-1}^{1} y\left(\tfrac{3}{2}y^2\right)dy = \tfrac{3}{2}\int_{-1}^{1} y^3 dy = \tfrac{3}{2}\left[\tfrac{y^4}{4}\right]_{-1}^{1} = \tfrac{3}{2}\left(\tfrac{1}{4}-\tfrac{1}{4}\right) = 0}$$

$$\boldsymbol{F(y) = \int_{-1}^{y} \tfrac{3}{2}w^2 dw = \tfrac{3}{2}\left[\tfrac{w^3}{3}\right]_{-1}^{y} = \tfrac{1}{2}(y^3+1);\ -1 < y < 1}$$

$$\boldsymbol{P(Y>0) = 1 - F(0) = 1 - \tfrac{1}{2} = \tfrac{1}{2}}$$

$$\boldsymbol{F(q) = 0.25 \to \frac{1}{2}(q^3+1) = 0.25 \to q^3 + 1 = 0.5 \to q^3 = -0.5 \to q = \sqrt[3]{-0.5}}$$